Do Online Portfolio Selection Algorithms Really Work?
－Critical Evaluation of Machine Learning and Heuristic Techniques－

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Abstract

Two new portfolio reconstruction algorithms, AntiCor (Borodin et al., 2004) and OLMAR (Li and Hoi, 2013), have been proposed and claimed that they significantly outperform market index by efficiently extracting mean-reverting phenomena. However, the datasets used in those studies did not cover well the collapse of an IT bubble economy (2000~2003) or Lehmann shock in 2008, and dealt with only a few blue chips. Therefore, it is necessary to point out the drawbacks of the above mentioned algorithms from the standpoints of practical use. For this purpose, the authors firstly examine what kind of datasets one should use and how each algorithm needs to be evaluated. Then their robustness is studied for the data of 60 – 430 brands in the U.S. market. In addition, maximum drawdown and asset power derived by pertinent transaction costs are evaluated and issues for practical use are investigated.

Keyword: Finance, Portfolio Selection, Stock Market, Machine Learning